JAVASCRIPT ASSIGNMENT : 09

Module 9: Strings and Numbers – Multiple Assignment Questions

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Section 1: String Methods

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1. Reverse a string manually

Create function that reverses given string without using built in reverse method.
Hint: Loop from last character to first.

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2. Check if string is palindrome

Take string and check whether it reads same forward and backward.
Hint: Compare original and reversed string.

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3. Count vowels in string

Take string and count total vowels present.
Hint: Loop through each character.

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4. Convert string to uppercase and lowercase

Create program that converts input string to uppercase and lowercase.
Hint: Use string methods.

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5. Find length of string without using length property

Create program to count characters manually.
Hint: Use loop and counter.

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6. Remove extra spaces from string

Create program that removes extra spaces between words.
Hint: Use trim and split logic.

7. Replace specific word in sentence

Replace word JavaScript with JS in sentence.
Hint: Use replace method.

8. Check if string contains specific word

Create program to check if string contains specific substring.
Hint: Use includes method.

9. Extract substring from string

Create program to extract specific part of string.
Hint: Use slice or substring.

10. Count number of words in sentence

Create program to count total words.
Hint: Split string by space.

Section 2: Template Literals

11. Create greeting message using template literals

Create message using name and age variables.
Hint: Use backticks and variable interpolation.

12. Create multi line string using template literals

Create formatted paragraph using template literals.
Hint: Avoid concatenation.

13. Create invoice generator

Create bill message using product name quantity and price using template literals.
Hint: Embed expressions inside template.

Section 3: Number Methods

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14. Convert string to number

Take numeric string and convert into number type.
Hint: Use number conversion method.

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15. Format number to fixed decimal places

Create program to round number to two decimal places.
Hint: Use toFixed method.

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16. Check if value is integer

Create program to check whether given value is integer.
Hint: Use number method.

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17. Parse floating number from string

Extract number from string like 45.67px.
Hint: Use parsing method.

Section 4: Math Object

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18. Generate random number between 1 and 100

Create program that generates random number.
Hint: Use random and floor methods.

19. Find maximum and minimum from numbers

Create program to find max and min values.
Hint: Use Math object methods.

20. Round number using different Math methods

Create program demonstrating round floor and ceil.
Hint: Compare outputs.

21. Calculate square root and power

Create program to calculate square root and power of number.
Hint: Use Math functions.

Section 5: Real World Logical Questions

22. Create password strength checker

Check if password length is greater than 8 and contains number.
Hint: Use string methods and condition.

23. Create email validation checker

Check if email contains at and dot symbols.
Hint: Use includes and indexOf.

24. Create OTP generator

Generate random 4 digit OTP.
Hint: Use random and floor logic.

25. Format currency output

Convert number into formatted currency string.
Hint: Use toFixed and template literals.

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26. Create number guessing game

Generate random number and allow user to guess.
Hint: Combine Math and conditions.

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27. Find longest word in sentence

Take sentence and return longest word.
Hint: Split and compare lengths.

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28. Count character frequency in string

Count how many times each character appears.
Hint: Use object for counting.

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29. Remove duplicate characters from string

Create program to remove duplicate letters.
Hint: Use loop and condition.

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30. Capitalize first letter of each word

Convert sentence into title case.
Hint: Split sentence and modify each word.